Internship Task Submission Report

C/C++ Programming Track — Core Task Evaluation

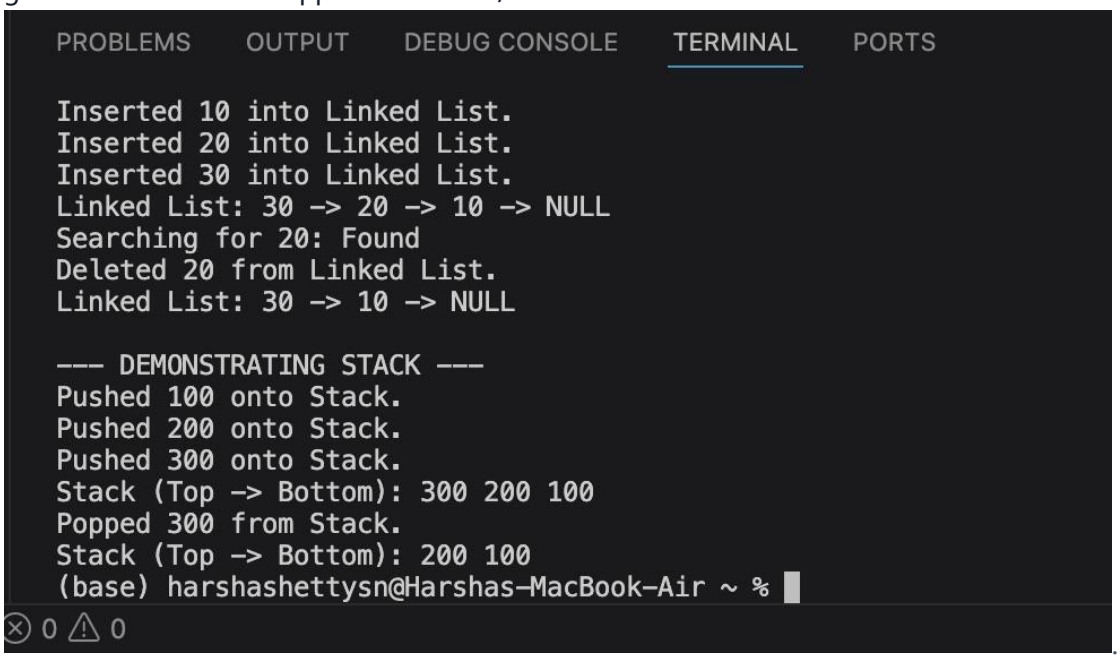
Developer Profile: Harsha1692006
Compiler Tools: Apple Clang / GCC toolchain (g++)
0Repository Link: https://github.com/Harsha1692006/InternSpark-Tasks

Task 1: Data Structures Implementation

Goal: Implement core fundamental data structures from scratch using pure C/C++ memory management,
avoiding external libraries or built-in collection primitives.

• **Selected Implementations:** Singly Linked List & Pointer-based Stack.
• **Key Methods Included:** Head insertion, value-targeted pointer node removals, linear element lookups,
and explicit stack push/pop mutations.
• **Time Complexities Documented:** Insertion/Push: *O(1)*, Search/Remove: *O(n)*.

g++ -std=c++17 main.cpp -o main && ./main

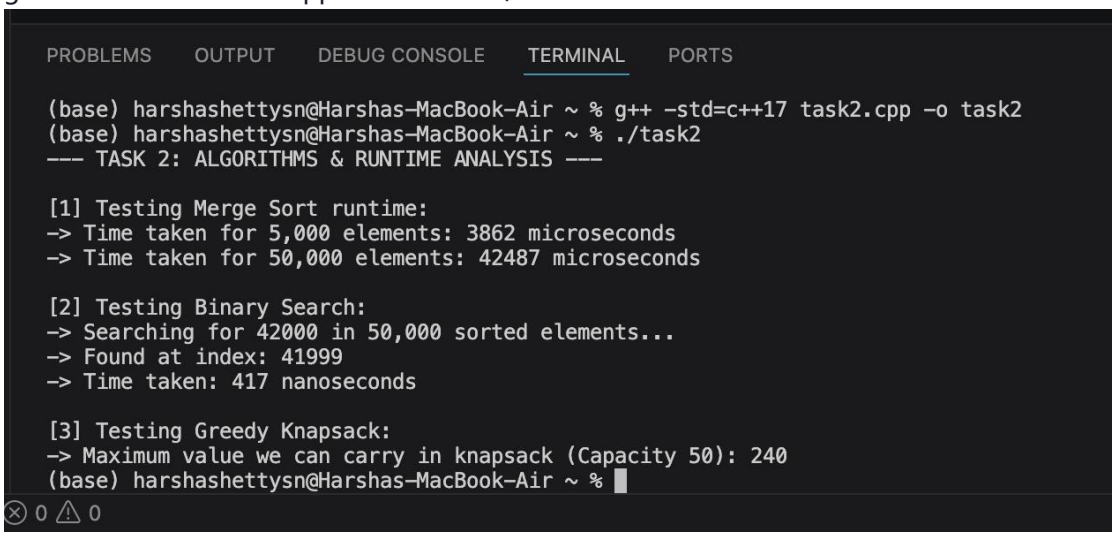


Task 2: Algorithms & Problem Solving

Goal: Implement standard search, sort, and greedy strategy functions alongside explicit code execution
tracking to measure exact CPU runtime overhead variations across varying array densities.

• **Selected Frameworks:** Divide-and-conquer Merge Sort, Binary Search tree paths, and a Greedy
fractional knapsack algorithm.
• **Measurement System:** Leveraged the native <chrono> structural library to isolate accurate metrics
across small scale inputs versus massive arrays.

g++ -std=c++17 task2.cpp -o task2 && ./task2

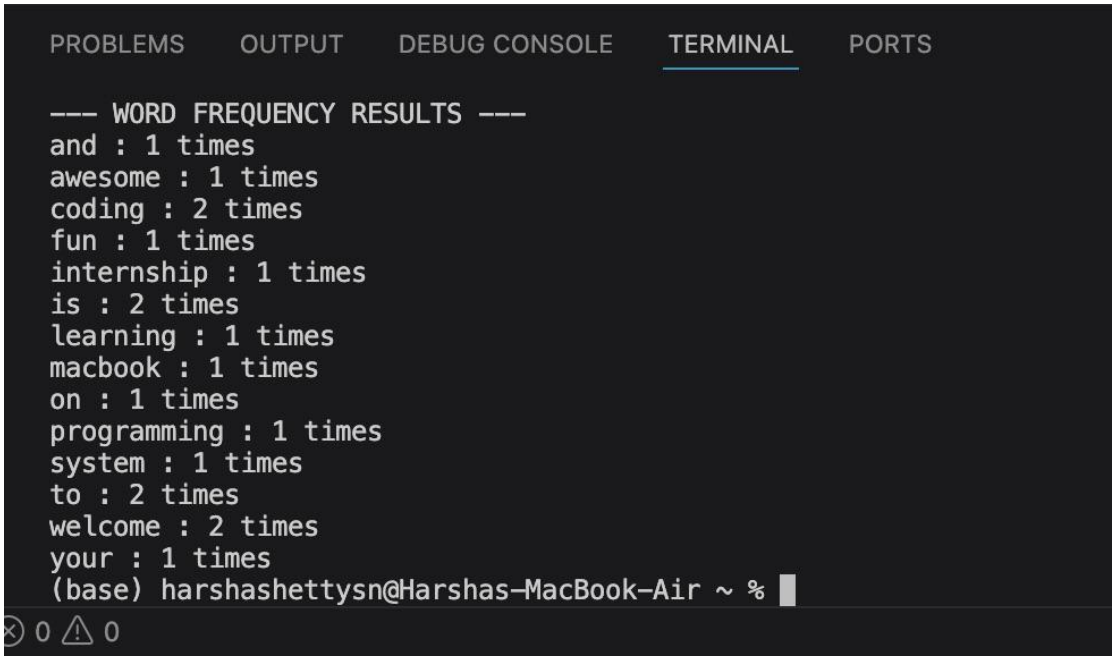


Task 3: System Programming & File I/O

Goal: Design an atomic system file routine mapping native character reading operations, standard text
streams, error check safety barriers, and token maps.

• **Architecture Details:** Programatically structures an external sample.txt log output block, establishes a
robust ifstream parse channel, normalizes character casings, and lists cumulative occurrence rates via
a lexical key-sorted map framework.

g++ -std=c++17 task3.cpp -o task3 && ./task3


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
--- WORD FREQUENCY RESULTS ---
and : 1 times
awesome : 1 times
coding : 2 times
fun : 1 times
internship : 1 times
is : 2 times
learning : 1 times
macbook : 1 times
on : 1 times
programming : 1 times
system : 1 times
to : 2 times
welcome : 2 times
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(base) harshashettysn@Harshas-MacBook-Air ~ %
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